Lecture 14 Tutorial 2 Scribe

Tutorial_2.pdf
PDF

Prompt for Lecture 14 (Tutorial 2):

"Generate a lecture scribe suitable for exam preparation based ONLY on the provided context (Tutorial 2 / Lecture 14 document). The target audience is a student preparing for a closed-notes exam.

Instructions:

Reason Step-by-Step: While writing the scribe, explicitly explain the reasoning behind the solutions for each tutorial question exactly as presented in the provided context. Include this reasoning directly in the scribe text to help the student understand 'why' and 'how' the problem is solved.

Faithfulness: Include notations, equations, and worked solutions exactly as they appear in the document. Do not hallucinate or add outside information.

Logical Flow: Make the logical dependencies between mathematical steps clear for each problem.

Structure: Organize the content with clear headings corresponding to each Tutorial Question (e.g., Question 1, Question 2, etc.).

Goal: The output should allow a student to understand the tutorial problems, reconstruct the mathematical steps, and follow the solutions step-by-step."

Below is a **Lecture 14 (Tutorial-2) Exam-Ready Scribe** based **only** on the provided document .

The goal is **step-by-step reasoning**, faithful to the questions, with clear logical flow so you can reconstruct every result in a closed-notes exam.

📘 CSE 400 – Fundamentals of Probability in Computing

Lecture 14 – Tutorial 2 Scribe (Exam Preparation)

Question 1

> A point is chosen at random on a line segment of length L. Interpret this statement, and find the probability that the ratio of the shorter to the longer segment is less than 1/4.

✅ Step 1: Interpret "chosen at random"

A point chosen uniformly on a segment of length L means:

$$X \sim \text{Uniform}(0, L)$$

The point divides the segment into:

- Left part = X
- Right part = $L - X$

✅ Step 2: Ratio condition

We want:

$$\frac{\min(X, L - X)}{\max(X, L - X)} < \frac{1}{4}$$

Because of symmetry, consider two cases:

Case 1: $X \leqslant L/2$

Shorter = X, Longer = $L - X$

$$\frac{X}{L-X} < \frac{1}{4}$$

Solve:

$$4X < L - X$$

$$5X < L$$

$$X < L/5$$

Case 2: $X > L/2$

By symmetry:

$$L - X < L/5$$

$$X > 4L/5$$

✅ Step 3: Compute probability

Since X is Uniform(0,L):

$$P = \frac{L/5}{L} + \frac{L/5}{L}$$
$$= \frac{2}{5}$$

✅ Final Answer:

$$\boxed{\frac{2}{5}}$$

Question 2

White region: $N(4, 4)$
Black region: $N(6, 9)$
Black fraction = α

Observed reading = 5

✅ Step 1: Understand error equality condition

We want:

Probability of error when deciding white

Probability of error when deciding **black**

Since observation is 5:

Error symmetry condition becomes:

$$\alpha f_B(5) = (1 - \alpha) f_W(5)$$

✅ Step 2: Write densities

White:

$$f_W(5) = \frac{1}{\sqrt{2\pi \cdot 4}} e^{-(5-4)^2/(2\cdot 4)}$$

Black:

$$f_B(5) = \frac{1}{\sqrt{2\pi \cdot 9}} e^{-(5-6)^2/(2\cdot 9)}$$

Note:

$$(5 - 4)^2 = 1$$

$$(5 - 6)^2 = 1$$

✅ Step 3: Solve for α

$$\alpha f_B(5) = (1 - \alpha) f_W(5)$$

$$\alpha = \frac{f_W(5)}{f_W(5) + f_B(5)}$$

That is the required value.

Question 3

(a) Uniform(0,A)

We minimize:

$$E[\,|X - a|\,]$$

For uniform distribution, expected absolute deviation minimized at **median**.

Uniform(0,A) median:

$$a = A/2$$

(b) Exponential(λ)

We minimize:

$$E[\,|X - a|\,]$$

For any continuous distribution, minimized at **median**.

Exponential CDF:

$$F(x) = 1 - e^{-\lambda x}$$

Set median:

$$1 - e^{-\lambda m} = 1/2$$

$$e^{-\lambda m} = 1/2$$

$$m = \frac{\ln 2}{\lambda}$$

Question 4

Battery life = mixture of exponentials.

We want:

$$P(X > t + s \mid X > t)$$

✅ Step 1: Conditional probability formula

$$= \frac{P(X > t+s)}{P(X > t)}$$

Since mixture:

$$P(X > t) = p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}$$

$$P(X > t+s) = p_1 e^{-\lambda_1 (t+s)} + p_2 e^{-\lambda_2 (t+s)}$$

✅ Final Answer

$$\frac{p_1 e^{-\lambda_1 (t+s)} + p_2 e^{-\lambda_2 (t+s)}}{p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}}$$

Question 5

Given:

$$X \sim Poisson(\lambda_1)$$

$$Y \sim Poisson(\lambda_2)$$

Independent.

Total requests:

$$X + Y \sim Poisson(\lambda_1 + \lambda_2)$$

Conditional distribution

Property:

Given total = n,

$$X \mid (X + Y = n) \sim Binomial(n, \frac{\lambda_1}{\lambda_1 + \lambda_2})$$

Final answer:

$$P(X = k \mid X + Y = n) = \binom{n}{k} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2}\right)^k \left(\frac{\lambda_2}{\lambda_1 + \lambda_2}\right)^{n-k}$$

Question 6

$X \sim Uniform[-2, 2]$

Given $Y = g(X)$ (piecewise graph).

Strategy:

1. Identify mapping intervals
2. Use transformation technique
3. For monotonic region:

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|$$

where $x = g^{-1}(y)$

Then:

$$F_Y(y) = \int_{-\infty}^{y} f_Y(u) du$$

Question 7

Stock model:

After 1000 periods:

Number of ups:

$$S_n \sim Binomial(1000, 0.52)$$

Price:

$$S_0 u^k d^{1000-k}$$

Want:

$$u^k d^{1000-k} \geqslant 1.3$$

Take log:

$$k \ln u + (1000 - k) \ln d \geqslant \ln 1.3$$

Solve threshold k.

Use Normal approximation:

$$\mu = 1000(0.52)$$

$$\sigma^2 = 1000(0.52)(0.48)$$

Standardize and compute probability.

Question 8

$X \sim Exponential(\lambda = 1/2)$

(a)

$$P(X > 2) = e^{-(1/2)2} = e^{-1}$$

(b) Memoryless property

$$P(X > 10 \mid X > 9) = P(X > 1) = e^{-(1/2)1} = e^{-1/2}$$

Question 9

$$Z = \sqrt{X^2 + Y^2}$$

Use transformation to polar coordinates:

Joint density:

$$f_{X,Y}(x, y) = f_X(x) f_Y(y)$$

In polar:

$$f_Z(z) = \int_0^{2\pi} f_X(z \cos \theta) f_Y(z \sin \theta) z d\theta$$

Question 10

$$M_n = \max(X_1, \ldots, X_n)$$

$$M_{n+1} = \max(M_n, X_{n+1})$$

Step 1

$$P(M_n \leqslant x) = F(x)^n$$

$$P(M_{n+1} \leqslant y) = F(y)^{n+1}$$

Joint:

For $x \leqslant y$,

$$P(M_n \leqslant x, M_{n+1} \leqslant y) = F(x)^n F(y)$$

🎯 Final Advice for Exam

Most important concepts tested:

- Median minimizes $E \mid X - a \mid$
- Mixture distributions
- Conditional Poisson → Binomial
- Exponential memoryless property
- Transformations
- Normal approximation to Binomial
- Order statistics

If you want, I can now convert this into **ultra-compact 2-page revision sheet** for last-day preparation.

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Below is a **Lecture 14 (Tutorial-2) Exam Preparation Scribe**, written strictly from the provided document .

The goal is to reconstruct each solution step-by-step so that you can reproduce every argument in a closed-notes exam.

CSE 400: Fundamentals of Probability in Computing

Question 1

> A point is chosen at random on a line segment of length L. Interpret this statement, and find the probability that the ratio of the shorter to the longer segment is less than $1/4$.

Step 1: Interpret "chosen at random"

Choosing a point at random on a segment of length L means the location X is:

$$X \sim \text{Uniform}(0, L)$$

So every point between 0 and L is equally likely.

The point divides the segment into:

- Left part: X
- Right part: $L - X$

The shorter part is:

$$\min(X, L - X)$$

The longer part is:

$$\max(X, L - X)$$

Step 2: Express the condition

We want:

$$\frac{\text{shorter}}{\text{longer}} < \frac{1}{4}$$

Because of symmetry, analyze two cases.

Case 1: $X \leqslant L/2$

Shorter = X, longer = $L - X$

$$\frac{X}{L - X} < \frac{1}{4}$$

Multiply both sides by $4(L - X)$:

$$4X < L - X$$

$$5X < L$$

$$X < \frac{L}{5}$$

Case 2: $X > L/2$

Now shorter = $L - X$

$$\frac{L - X}{X} < \frac{1}{4}$$

This gives:

$$L - X < \frac{X}{4}$$

$$4L - 4X < X$$

$$4L < 5X$$

$$X > \frac{4L}{5}$$

Step 3: Compute probability

The favorable region is:

$$X < \frac{L}{5} \quad \text{or} \quad X > \frac{4L}{5}$$

Each interval has length $L/5$.

Since total length is L,

$$P = \frac{L/5 + L/5}{L} = \frac{2L/5}{L} = \frac{2}{5}$$

Final Answer

$$\boxed{\frac{2}{5}}$$

Question 2

White region:

$$N(4, 4)$$

Black region:

$$N(6, 9)$$

Fraction of image black = α

Observed reading = 5.

We want the value of α such that probability of error is the same, whether concluding black or white.

Step 1: Equal error condition

At observation $x = 5$, equal probability of error means:

$$\alpha f_B(5) = (1 - \alpha) f_W(5)$$

This balances weighted likelihoods.

Step 2: Write densities

Normal density:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/(2\sigma^2)}$$

For white:

$$f_W(5) = \frac{1}{\sqrt{2\pi \cdot 4}} e^{-(5-4)^2/(2\cdot 4)}$$

For black:

$$f_B(5) = \frac{1}{\sqrt{2\pi \cdot 9}} e^{-(5-6)^2/(2\cdot 9)}$$

Since:

$$(5-4)^2 = 1, \qquad (5-6)^2 = 1$$

Step 3: Solve for α

$$\alpha f_B(5) = (1-\alpha) f_W(5)$$

$$\alpha f_B(5) = f_W(5) - \alpha f_W(5)$$

$$\alpha(f_B(5) + f_W(5)) = f_W(5)$$

$$\alpha = \frac{f_W(5)}{f_W(5) + f_B(5)}$$

Question 3

(a) Uniform(0, A)

Minimize:

$$E[\,|X-a|\,]$$

For absolute deviation, minimum occurs at the median.

Uniform(0,A) median:

$$\boxed{a = A/2}$$

(b) Exponential(λ)

Again minimize:

$$E[\,|X-a|\,]$$

Minimum occurs at median.

Exponential CDF:

$$F(x) = 1 - e^{-\lambda x}$$

Set $F(m) = 1/2$:

$$1 - e^{-\lambda m} = 1/2$$

$$e^{-\lambda m} = 1/2$$

$$m = \frac{\ln 2}{\lambda}$$

Question 4

Two types of batteries:

Type i: exponential(λ_i)
Probability p_i

We want:

$$P(X > t + s \mid X > t)$$

Step 1: Conditional probability

$$= \frac{P(X > t + s)}{P(X > t)}$$

Since mixture:

$$P(X > t) = p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}$$

$$P(X > t + s) = p_1 e^{-\lambda_1 (t+s)} + p_2 e^{-\lambda_2 (t+s)}$$

Final Answer

$$\frac{p_1 e^{-\lambda_1 (t+s)} + p_2 e^{-\lambda_2 (t+s)}}{p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}}$$

Question 5

$$X \sim Poisson(\lambda_1), \quad Y \sim Poisson(\lambda_2)$$

Independent.

Total:

$$X + Y \sim Poisson(\lambda_1 + \lambda_2)$$

We want:

$$P(X = k \mid X + Y = n)$$

Step 1: Conditional probability formula

$$P(X = k \mid X + Y = n) = \frac{P(X = k, Y = n - k)}{P(X + Y = n)}$$

Using Poisson independence:

$$= \frac{\frac{e^{-\lambda_1}\lambda_1^k}{k!} \frac{e^{-\lambda_2}\lambda_2^{n-k}}{(n-k)!}}{\frac{e^{-(\lambda_1+\lambda_2)}(\lambda_1+\lambda_2)^n}{n!}}$$

Cancel exponentials.

Rearrange factorials:

$$= \binom{n}{k} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2}\right)^k \left(\frac{\lambda_2}{\lambda_1 + \lambda_2}\right)^{n-k}$$

Question 6

$X \sim Uniform[-2, 2]$
$Y = g(X)$

Strategy

To find $F_Y(y)$:

$$F_Y(y) = P(Y \leqslant y) = P(g(X) \leqslant y)$$

Express this in terms of ranges of X.

To find density:

If transformation is monotonic:

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|$$

where $x = g^{-1}(y)$.

Then integrate to obtain CDF.

Question 7

After 1000 periods:

Number of up movements:

$$S \sim Binomial(1000, 0.52)$$

Price becomes:

$$su^k d^{1000-k}$$

We require:

$$u^k d^{1000-k} \geqslant 1.3$$

Taking log:

$$k \ln u + (1000 - k) \ln d \geqslant \ln 1.3$$

Solve inequality for k.

Approximate binomial by normal:

$$\mu = 1000(0.52), \quad \sigma^2 = 1000(0.52)(0.48)$$

Standardize and compute tail probability.

Question 8

$$X \sim Exponential(\lambda = 1/2)$$

(a)

$$P(X > 2) = e^{-(1/2)2} = e^{-1}$$

(b)

Memoryless property:

$$P(X > 10 \mid X > 9) = P(X > 1) = e^{-(1/2)1} = e^{-1/2}$$

Question 9

$$Z = \sqrt{X^2 + Y^2}$$

Joint density:

$$f_{X,Y}(x, y) = f_X(x) f_Y(y)$$

Use polar coordinates:

$$x = z\cos\theta, \quad y = z\sin\theta$$

Jacobian = z.

$$f_Z(z) = \int_0^{2\pi} f_X(z\cos\theta) f_Y(z\sin\theta) z \, d\theta$$

Question 10

$$M_n = \max(X_1, \dots, X_n)$$

$$M_{n+1} = \max(M_n, X_{n+1})$$

Step 1

$$P(M_n \leqslant x) = F(x)^n$$

$$P(M_{n+1} \leqslant y) = F(y)^{n+1}$$

Joint distribution

For $x \leqslant y$:

$$P(M_n \leqslant x, M_{n+1} \leqslant y) = F(x)^n F(y)$$

Final Exam Strategy

From this tutorial, you must be able to:

- Work with absolute values and medians
- Handle mixture distributions
- Perform conditional probability with Poisson
- Use exponential memorylessness
- Apply transformations
- Approximate binomial with normal
- Work with maxima and order statistics

If you want, I can now create a **condensed 1-page ultra-revision sheet** for fast exam review.

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Lecture 14 – Tutorial 2 Scribe

Question 1

> A point is chosen at random on a line segment of length L. Interpret this statement, and find the probability that the ratio of the shorter to the longer segment is less than $1/4$.

Step 1: Interpret “chosen at random”

A point chosen at random on a segment of length L means the point location X is uniformly distributed on $(0, L)$:

$$X \sim \text{Uniform}(0, L)$$

Thus, every subinterval of equal length is equally likely.

The point divides the segment into:

$$X \quad \text{and} \quad L - X$$

The shorter piece:

$$\min(X, L - X)$$

The longer piece:

$$\max(X, L - X)$$

Step 2: Express the required inequality

We want:

$$\frac{\min(X, L - X)}{\max(X, L - X)} < \frac{1}{4}$$

Because the distribution is symmetric about $L/2$, we analyze two regions.

Case 1: $0 < X \leqslant L/2$

Shorter segment = X
Longer segment = $L - X$

Condition:

$$\frac{X}{L - X} < \frac{1}{4}$$

Multiply both sides by $4(L - X)$:

$$4X < L - X$$

$$5X < L$$

$$X < \frac{L}{5}$$

Case 2: $L/2 < X < L$

Shorter segment = $L - X$

Condition:

$$\frac{L - X}{X} < \frac{1}{4}$$

Multiply:

$$4(L - X) < X$$

$$4L - 4X < X$$

$$4L < 5X$$

$$X > \frac{4L}{5}$$

Step 3: Compute probability

Thus favorable region:

$$X < \frac{L}{5} \quad \text{or} \quad X > \frac{4L}{5}$$

Each interval has length $L/5$.

Since total length is L:

$$P = \frac{L/5 + L/5}{L} = \frac{2L/5}{L} = \frac{2}{5}$$

Final Answer:

$$\boxed{\frac{2}{5}}$$

Question 2

White region reading:

$$N(4, 4)$$

Black region reading:

$$N(6, 9)$$

Fraction black = α

Observed reading = 5.

We must determine α so that probability of error is equal regardless of concluding black or white.

Step 1: Equal error condition

At observation $x = 5$, equal probability of misclassification requires:

$$\alpha f_B(5) = (1 - \alpha) f_W(5)$$

This balances posterior weighted likelihoods.

Step 2: Write normal densities

General normal density:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/(2\sigma^2)}$$

White:

$$f_W(5) = \frac{1}{\sqrt{2\pi \cdot 4}} e^{-(5-4)^2/(2\cdot 4)}$$

Black:

$$f_B(5) = \frac{1}{\sqrt{2\pi \cdot 9}} e^{-(5-6)^2/(2\cdot 9)}$$

Since:

$$(5-4)^2 = 1, \quad (5-6)^2 = 1$$

Step 3: Solve for α

$$\alpha f_B(5) = (1-\alpha) f_W(5)$$

$$\alpha(f_B(5) + f_W(5)) = f_W(5)$$

$$\alpha = \frac{f_W(5)}{f_W(5) + f_B(5)}$$

Question 3

(a) Uniform(0,A)

We minimize:

$$E[\,|X-a|\,]$$

For absolute deviation, the minimizer is the median.

For $X \sim \text{Uniform}(0, A)$:

$$\boxed{a = A/2}$$

(b) Exponential(λ)

Minimize:

$$E[\,|X-a|\,]$$

Again minimized at the median.

For exponential:

$$F(x) = 1 - e^{-\lambda x}$$

Median m satisfies:

$$1 - e^{-\lambda m} = \frac{1}{2}$$

$$e^{-\lambda m} = \frac{1}{2}$$

$$m = \frac{\ln 2}{\lambda}$$

Question 4

Two battery types:

Type i lifetime ~ Exponential(λ_i)
Selected with probability p_i, with $p_1 + p_2 = 1$.

We want:

$$P(X > t + s \mid X > t)$$

Step 1: Use definition of conditional probability

$$= \frac{P(X > t + s)}{P(X > t)}$$

Because mixture:

$$P(X > t) = p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}$$

$$P(X > t + s) = p_1 e^{-\lambda_1 (t+s)} + p_2 e^{-\lambda_2 (t+s)}$$

Final Answer:

$$\frac{p_1 e^{-\lambda_1 (t+s)} + p_2 e^{-\lambda_2 (t+s)}}{p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}}$$

Question 5

$$X \sim \text{Poisson}(\lambda_1)$$

$$Y \sim \text{Poisson}(\lambda_2)$$

Independent.

Find:

$$P(X = k \mid X + Y = n)$$

Step 1: Use conditional definition

$$P(X = k \mid X + Y = n) = \frac{P(X = k, Y = n - k)}{P(X + Y = n)}$$

Step 2: Substitute Poisson pmf

$$= \frac{\frac{e^{-\lambda_1}\lambda_1^k}{k!}\frac{e^{-\lambda_2}\lambda_2^{n-k}}{(n-k)!}}{\frac{e^{-(\lambda_1+\lambda_2)}(\lambda_1+\lambda_2)^n}{n!}}$$

Cancel exponentials and rearrange:

$$= \binom{n}{k}\left(\frac{\lambda_1}{\lambda_1 + \lambda_2}\right)^k \left(\frac{\lambda_2}{\lambda_1 + \lambda_2}\right)^{n-k}$$

Question 6

$X \sim \text{Uniform}[-2, 2]$
$Y = g(X)$ (as given by graph).

Step 1: CDF

$$F_Y(y) = P(Y \leqslant y) = P(g(X) \leqslant y)$$

Express inequality in terms of X.
Determine intervals of X satisfying it.

Step 2: Density

If g is monotonic in a region:

$$f_Y(y) = f_X(x)\left|\frac{dx}{dy}\right| \quad \text{where } x = g^{-1}(y)$$

Compute for each monotonic segment and sum.

Question 7

After 1000 periods:

Number of upward moves:

$$S \sim \text{Binomial}(1000, 0.52)$$

Price:

$$u^k d^{1000-k}$$

We require:

$$u^k d^{1000-k} \geqslant 1.3$$

Take logarithm:

$$k \ln u + (1000 - k) \ln d \geqslant \ln 1.3$$

Solve for critical k.

Approximate binomial by normal:

$$\mu = 1000(0.52), \quad \sigma^2 = 1000(0.52)(0.48)$$

Standardize and compute tail probability.

Question 8

$$X \sim \text{Exponential}(\lambda = 1/2)$$

(a)

$$P(X > 2) = e^{-(1/2)2} = e^{-1}$$

(b)

Using memoryless property:

$$P(X > 10 \mid X > 9) = P(X > 1)$$
$$= e^{-(1/2)1} = e^{-1/2}$$

Question 9

$$Z = \sqrt{X^2 + Y^2}$$

Since independence:

$$f_{X,Y}(x, y) = f_X(x) f_Y(y)$$

Change to polar:

$$x = z\cos\theta, \quad y = z\sin\theta$$

Jacobian $= z$.

$$f_Z(z) = \int_0^{2\pi} f_X(z\cos\theta) f_Y(z\sin\theta) z \, d\theta$$

Question 10

$$M_n = \max(X_1, \dots, X_n)$$
$$M_{n+1} = \max(M_n, X_{n+1})$$

Step 1

$$P(M_n \leqslant x) = F(x)^n$$
$$P(M_{n+1} \leqslant y) = F(y)^{n+1}$$

Joint distribution (for $x \leqslant y$)

$$P(M_n \leqslant x, M_{n+1} \leqslant y) = F(x)^n F(y)$$

End of Tutorial-2 Lecture 14 Scribe

This version contains all logical steps required to reconstruct each derivation for a closed-notes examination directly from the provided tutorial document .